

Project Design Phase-II Technology Stack (Architecture & Stack)

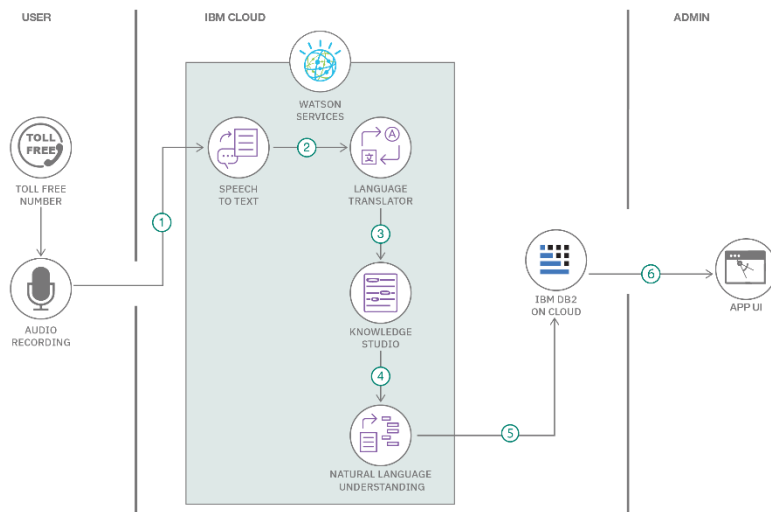
Date	31 January 3035
Team ID	LTVIP2025TMID45199
Project Name	Enchanted Wings: Marvels of Butterfly Species
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	HTML, CSS, JavaScript / Angular Js / React Js etc.
2.	Application Logic-1	Logic for a process in the application	Java / Python
3.	Application Logic-2	Logic for a process in the application	IBM Watson STT service
4.	Application Logic-3	Logic for a process in the application	IBM Watson Assistant
5.	Database	Data Type, Configurations etc.	MySQL, NoSQL, etc.
6.	Cloud Database	Database Service on Cloud	IBM DB2, IBM Cloudant etc.
7.	File Storage	File storage requirements	IBM Block Storage or Other Storage Service or Local Filesystem
8.	External API-1	Purpose of External API used in the application	IBM Weather API, etc.
9.	External API-2	Purpose of External API used in the application	Aadhar API, etc.
10.	Machine Learning Model	Purpose of Machine Learning Model	Object Recognition Model, etc.
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration :	Local, Cloud Foundry, Kubernetes, etc.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Technology of Opensource framework
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	e.g. SHA-256, Encryptions, IAM Controls, OWASP etc.

S.No	Characteristics	Description	Technology
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	Technology used
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	Technology used
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Technology used

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

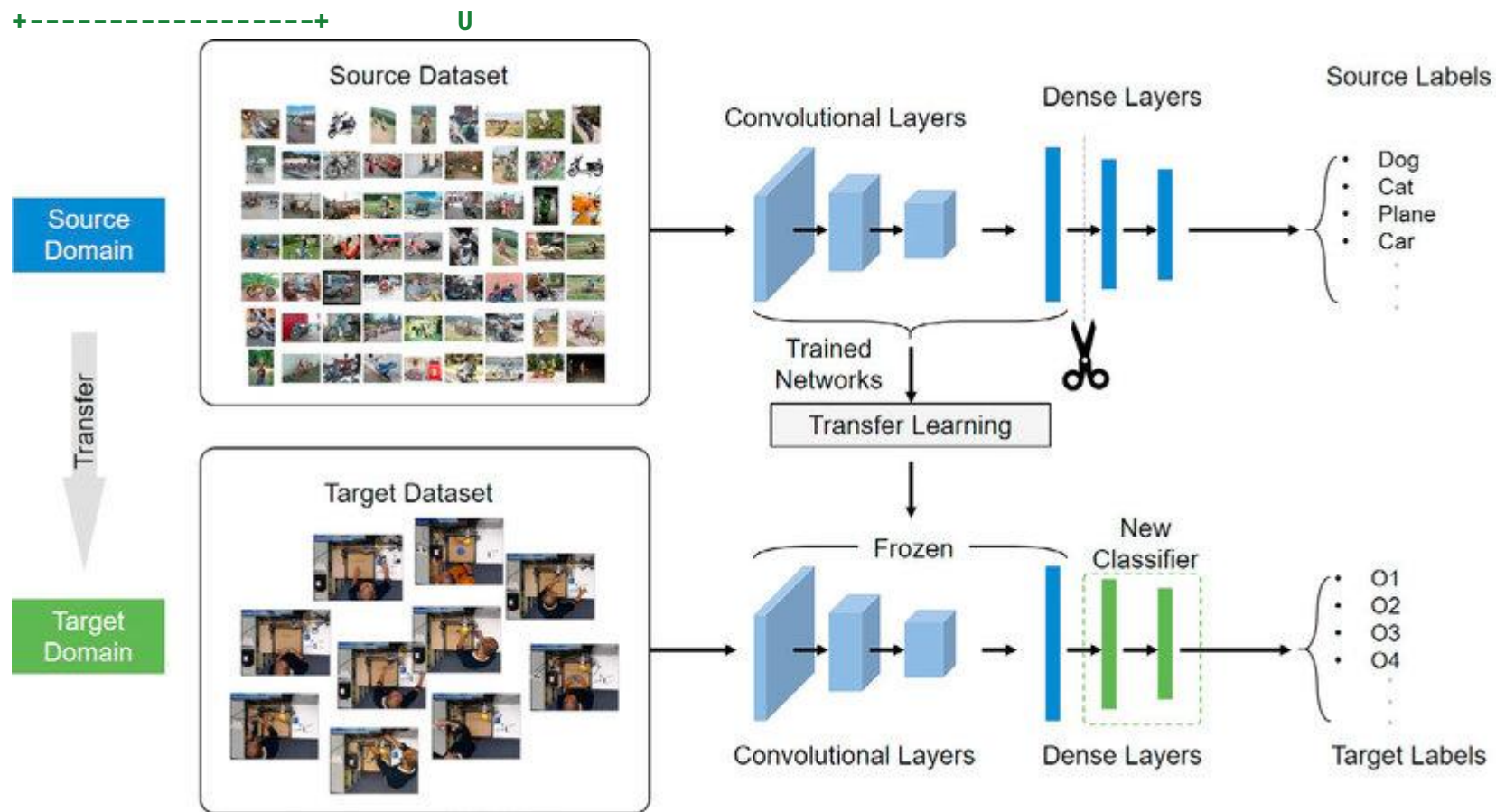
<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>

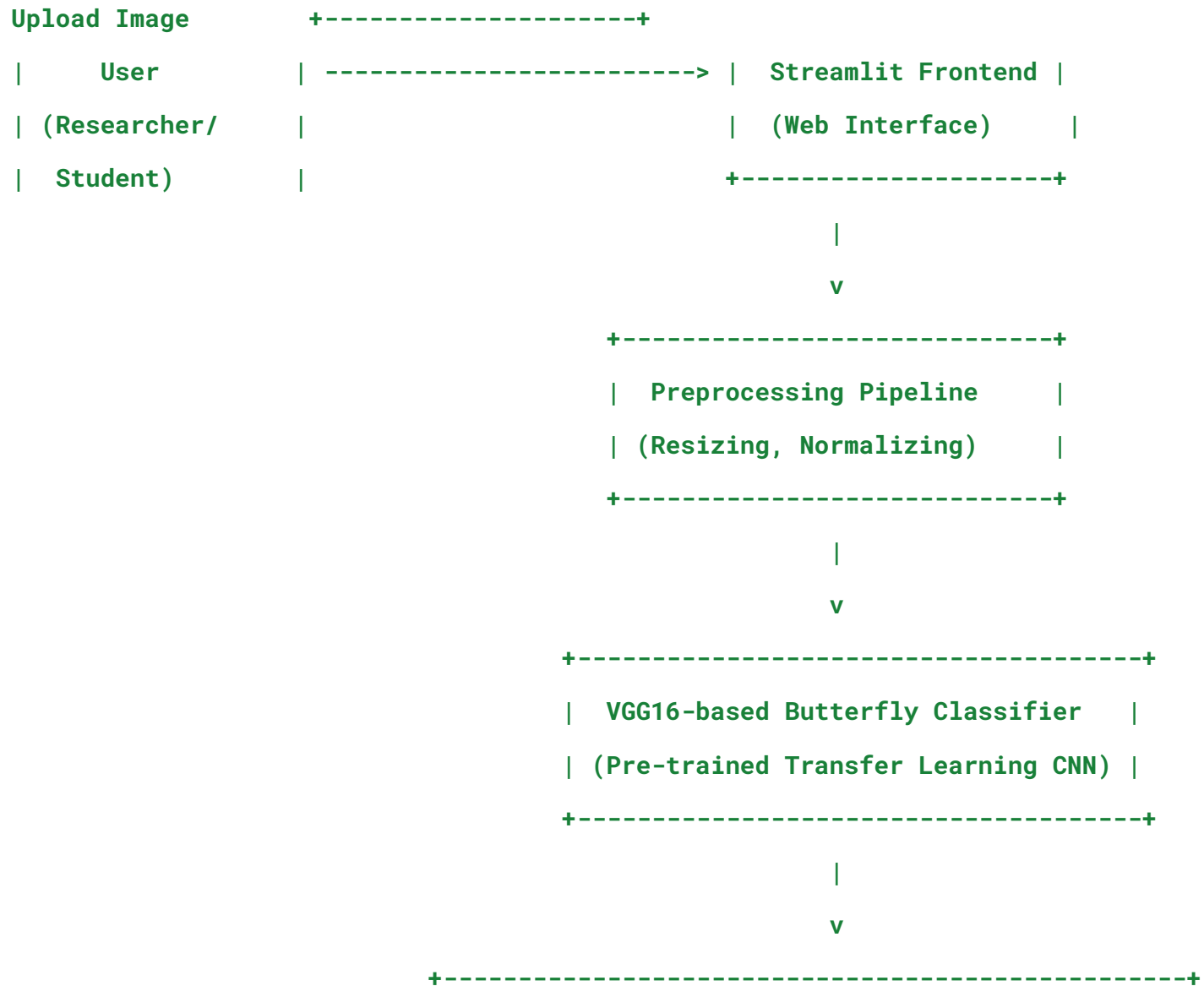


Technical Architecture for: *Enchanted Wings: Marvels of Butterfly Species*



Architecture Diagram Overview





```
| Display Prediction + Species Info + Charts      |
| (Confidence Score, Scientific Name, Distribution)|
+-----+
```

 **Table-1: Components & Technologies**

S.No	Component	Description	Technology
1	User Interface	Web interface for uploading butterfly images & displaying results	HTML, CSS, JavaScript, Streamlit
2	Application Logic-1	Model loading, prediction, result formatting	Python
3	Application Logic-2	Preprocessing pipeline (resizing, normalizing images)	OpenCV, NumPy
4	Application Logic-3	Visualization and plotting of training results	Matplotlib, Seaborn
5	Database	Store feedback, usage logs	SQLite / Firebase

6	Cloud Database	Optional storage of dataset and results for analysis	Google Firebase / AWS S3
7	File Storage	Image datasets and prediction logs	Local File System / Google Drive
8	External API-1	Pull butterfly information based on species name	Wikipedia API
9	External API-2	Optional geolocation metadata of image	Google Maps API (if used)
10	Machine Learning Model	Transfer learning-based classifier for butterfly species	VGG16 via TensorFlow / Keras
11	Infrastructure	Deployed as local demo or cloud Streamlit App	Localhost / Streamlit Cloud / Heroku

 **Table-2: Application Characteristics**

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Keras, TensorFlow, Streamlit, Scikit-learn, Matplotlib	Python Libraries

2	Security Implementations	Secure file upload, HTTPS enabled if hosted, access logs	SHA-256 (if login), HTTPS, Firebase Auth
3	Scalable Architecture	Modular logic, separated model + interface, support for re-training	3-tier architecture
4	Availability	Available via Streamlit Cloud with continuous deployment	Streamlit Sharing / Heroku
5	Performance	Pre-trained model reduces inference time; caching with @st.cache_resource	TensorFlow Lite (for optimization), CDN

✓ References

- <https://c4model.com/>
- <https://developer.ibm.com/patterns/>
- <https://aws.amazon.com/architecture>
- <https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>